

§5.1 Wrenches

Generalized force as a 6×1 column vector (or $\in \mathbb{R}^6$):

$$F = \begin{bmatrix} f \\ \tau \end{bmatrix} \quad \begin{array}{ll} f \in \mathbb{R}^3 & \text{linear component} \\ \tau \in \mathbb{R}^3 & \text{rotational component} \end{array}$$

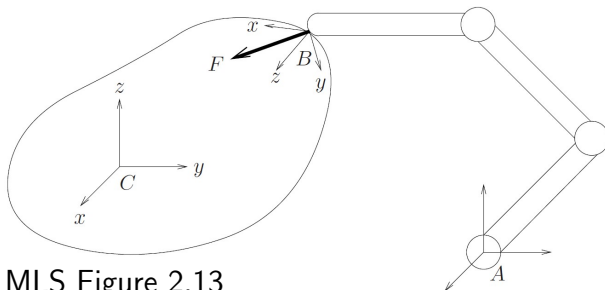
This force/moment pair is called a wrench.

Infinitesimal work done by wrench F_b acting at a body frame B origin, where the body is undergoing motion parameterized by $g_{ab}(t)$, frame A inertial:

$$\delta W = V_{ab}^{bT} F_b = v_{ab}^{bT} f_b + \omega_{ab}^T \tau_b$$

§5.1 Wrenches

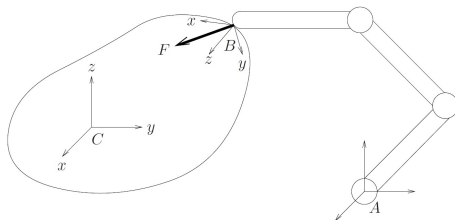
Two wrenches are said to be equivalent if they generate the same work for every possible rigid body motion.



MLS Figure 2.13

- ▶ Frame B : origin of contact frame; also a body frame
- ▶ Frame C : a body frame
- ▶ F_b a wrench applied to the body at the origin of B
- ▶ Wish to determine the equivalent wrench applied at the origin of C , F_c

§5.1 Wrenches



Let $g_{bc} = (R_{bc}, p_{bc})$ be the configuration of C relative to B
 B and C fixed to a single rigid body $\implies \dot{g}_{bc}$ a zero matrix

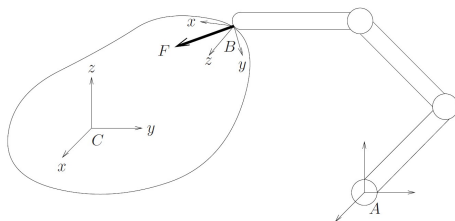
Eq. (2.53) p. 54, $V_{bc}^s = (\dot{g}_{bc} g_{bc}^{-1})^\vee = 0$

Proposition 2.14 p. 59, $V_{ac}^s = V_{ab}^s$

Eq. (2.55) p. 55, $V_{bc}^b = (g_{bc}^{-1} \dot{g}_{bc})^\vee = 0$

Proposition 2.15 p. 59, $V_{ac}^b = \text{Ad}_{g_{bc}^{-1}} V_{ab}^b \implies V_{ab}^b = \text{Ad}_{g_{bc}} V_{ac}^b$

§5.1 Wrenches



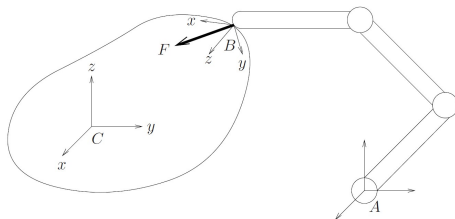
$$V_{ab}^b = \text{Ad}_{g_{bc}} V_{ac}^b$$

$$\delta W = V_{ac}^{bT} F_c = V_{ab}^{bT} F_b = (\text{Ad}_{g_{bc}} V_{ac}^b)^T F_b = V_{ac}^{bT} \text{Ad}_{g_{bc}}^T F_b$$

$$F_c = \text{Ad}_{g_{bc}}^T F_b \quad (2.65)$$

Transforms wrench applied at origin of B to equivalent wrench at origin C on same rigid body

§5.1 Wrenches



$$F_c = \text{Ad}_{g_{bc}}^T F_b \quad (2.65)$$

$$\begin{bmatrix} f_c \\ \tau_c \end{bmatrix} = \begin{bmatrix} R_{bc}^T & 0 \\ -R_{bc}^T \hat{p}_{bc} & R_{bc}^T \end{bmatrix} \begin{bmatrix} f_b \\ \tau_b \end{bmatrix} \quad (2.66)$$

Torque $-\hat{p}_{bc}f_b$?

§5.1 Wrenches

Several wrenches applied to a body: Net wrench constructed by determining equivalent wrench of each at a chosen point and then summing these equivalent wrench vectors

§5.1 Wrenches

For $g \in SE(3)$ describing the configuration of a rigid body, F^b and F^s are the body wrench and spatial wrench, respectively (p. 63).

$$F^b = \text{Ad}_g^T F^s \quad (2.67)$$

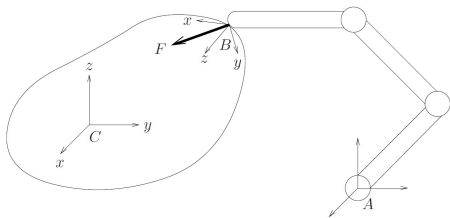
$F^s?$

(See top p. 63: $F_c = \text{Ad}_{g_{bc}}^T F_b$ (2.65) $\implies F_a = \text{Ad}_{g_{ba}}^T F_b$)

$$F_a = \text{Ad}_{g_{ba}}^T F_b$$

implies

$$F_b = \text{Ad}_{g_{ab}}^T F_a$$



§5.1 Wrenches

$$V^s = \text{Ad}_g V^b \quad (2.61)$$

$$F^b = \text{Ad}_g^T F^s \quad (2.67)$$

From eq. (2.61) follows

$$V^b = \text{Ad}_g^{-1} V^s$$

Therefore

$$\begin{aligned} \delta W &= V^{bT} F^b = (\text{Ad}_g^{-1} V^s)^T \text{Ad}_g^T F^s = V^{sT} \text{Ad}_g^{-T} \text{Ad}_g^T F^s \\ &= V^{sT} F^s \end{aligned}$$

§5.2 Screw coordinates of a wrench

With respect to spatial frame A , let S be a screw: axis $l = \{q + \lambda\omega : \lambda \in \mathbb{R}\}$, $\|\omega\| = 1$, pitch h , magnitude M .

Construct wrench: force magnitude M along l and torque hM both in direction ω . If $h = \infty$, pure torque $M\omega$.

$$\begin{aligned} F &= M \begin{bmatrix} \omega \\ -\hat{\omega}q + h\omega \end{bmatrix} & h \text{ finite} \\ F &= M \begin{bmatrix} 0 \\ \omega \end{bmatrix} & h = \infty \end{aligned} \tag{2.68}$$

For h finite, the moment of the offset force $M\omega$ about the origin of A is a cross product of the form $\vec{r} \times \vec{F}$; here expressed as $\hat{q}\omega M = -\hat{\omega}qM$.

§5.2 Screw coordinates of a wrench

Theorem 2.17 (Poinsot) Every collection of wrenches applied to a rigid body is equivalent to a force applied along a fixed axis plus a torque about the same axis.

Let $F = (f, \tau)$ be the net wrench applied to a rigid body.

Case 1: $f = 0$ (pure torque). Set $M = \|\tau\|$, $\omega = \tau/M$ and $h = \infty$. Eq. (2.68) verifies theorem for this case.

Case 2: $f \neq 0$. Set $M = \|f\|$, $\omega = f/M$. We must now solve

$$M(\hat{q}\omega + h\omega) = \tau$$

to determine q and h .

§5.2 Screw coordinates of a wrench

Suggested solution:

$$h = \frac{f^T \tau}{||f||^2} \quad \text{and} \quad q = \frac{\hat{f} \tau}{||f||^2}$$

While realizing $||f|| = M$, if this solution is substituted into

$$M(\hat{q}\omega + h\omega) ,$$

we find

$$M \left[\widehat{\left(\frac{\hat{f} \tau}{M^2} \right)} \omega + \frac{f^T \tau}{M^2} \omega \right] = (\widehat{\hat{\omega} \tau}) \omega + (\omega^T \tau) \omega ,$$

where $f/M = \omega$ was substituted.

§5.2 Screw coordinates of a wrench

That

$$(\widehat{\omega\tau})\omega + (\omega^T\tau)\omega = \tau$$

if $\|\omega\| = 1$ was used before and is proven in the MATLAB script file “ident4MLS_Theor2_17.pdf”, available on Moodle under the ”Notes” topic.

QED

§5.2 Screw coordinates of a wrench

Using Poincot's theorem, we define screw coordinates of a wrench $F = (f, \tau)$:

1. Pitch:

$$h = \frac{f^T \tau}{\|f\|^2} \quad (2.69)$$

ratio of angular force to linear torque. $f = 0 \Rightarrow h = \infty$.

2. Axis:

$$l = \begin{cases} \left\{ \frac{\hat{f} \tau}{\|f\|^2} + \lambda f : \lambda \in \mathbb{R} \right\}, & \text{if } f \neq 0 \\ \{0 + \lambda \tau : \lambda \in \mathbb{R}\}, & \text{if } f = 0 \end{cases} \quad (2.70)$$

§5.2 Screw coordinates of a wrench

3. Magnitude:

$$M = \begin{cases} ||f||, & \text{if } f \neq 0 \\ ||\tau||, & \text{if } f = 0 \end{cases} \quad (2.71)$$

net linear force if motion contains a linear component, net torque otherwise.